MBI802 DATABASE MANAGEMENT SYSTEMS

ER Diagram Tutorial

Entities, attributes and relationships in Chen's notation

Lecturer **Yasas Sri Wickramasinghe** Time **About 90 minutes**

You need **A pencil, an eraser and this handout**

Instructions

1. Read Part 1 and Part 2 before you start the tasks.
2. Complete the tasks in order. Tasks 1, 2 and 6 are simple. Tasks 3, 4 and 5 are moderate. Task 6 is a short extra on participation.
3. For each task, read the scenario, answer the questions in the working space, then draw your ER diagram on the next page.
4. Check your work with the separate answers document, one part at a time: entities, then attributes, then relationships.

Learning outcomes

After this tutorial you should be able to:

- identify the entities in a scenario
- list the attributes of each entity and choose the key attribute
- identify the relationships and their cardinality
- draw an ER diagram using Chen's notation
- use the correct naming conventions.

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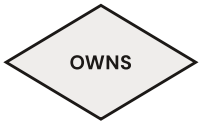
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PART 1

Chen's notation

An ER (entity relationship) diagram shows the data an organisation needs to store and how that data is connected. In this course we use Chen's notation, which uses four shapes.

Shape	Name	What it shows	How to find it in a scenario
	**Entity** Rectangle	A person, place, thing or event that the organisation stores data about. Each entity becomes a table.	Usually a noun, such as customer or course.
	**Attribute** Ellipse	One piece of information about an entity. Each attribute becomes a column.	Often follows "each customer has a…".
	**Key attribute** Ellipse, name underlined	An attribute whose value is different for every instance. It becomes the primary key.	Often an ID, a number or a code.
	**Relationship** Diamond	Shows how two entities are connected.	Usually a verb, such as owns, joins or teaches.

Lines connect the shapes. Each attribute is connected to its entity, and each relationship is connected to the two entities it links. The numbers 1, N and M on a relationship line show the cardinality (see Part 2).

Naming conventions

Each entity later becomes a table and each attribute becomes a column. Use these rules for every diagram.

Shape	Rule	Correct	Incorrect
Entity	Singular noun in capitals. Use _ between words.	`CUSTOMER, ID_CARD`	~~`Customers, customer`~~
Relationship	Verb in capitals. Use _ between words.	`OWNS, WORKS_IN`	~~`Ownership, works in`~~
Attribute	No spaces. Start each word with a capital letter.	`FirstName, DateOfBirth`	~~`first name, DOB, fname`~~
Key attribute	Same as an attribute, with the name underlined. Usually ends in ID, Number or Code.	**`CustomerID`**	~~`id, Cust_No`~~

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PART 2

How to draw an ER diagram

Follow these steps for each task.

1. **Read the scenario.** Read it once to understand it. Then read it again and mark the important words.
2. **Find the entities.** Look for nouns that the organisation stores data about. Do not include the organisation itself.
3. **List the attributes.** Write down the facts stored about each entity. Choose the key attribute: the one that is unique for each instance.
4. **Find the relationships.** Look for verbs that connect two entities. Write each one as ENTITY VERB ENTITY, for example CUSTOMER OWNS ACCOUNT.
5. **Work out the cardinality and draw.** Use the two questions below for each relationship. Draw the entities first, then the relationships, then the attributes.

Finding the cardinality

For each relationship, ask one question from each side. For example:

One CUSTOMER can own how many ACCOUNTs?	**Many**
One ACCOUNT is owned by how many CUSTOMERs?	**One**

Then use the answers to choose the cardinality:

Both answers are “one”	**1:1**
One answer is “one” and the other is “many”	**1:N**
Both answers are “many”	**M:N**

Where the numbers go. Write each number next to the entity it counts. A customer has many accounts, so N goes next to ACCOUNT. An account has one customer, so 1 goes next to CUSTOMER.

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TASK 1 Simple · Bank · About 10 minutes

Customers and accounts

Two entities and one relationship. The other tasks follow the same steps.

Scenario

> Mere runs a small community bank in Hamilton. She keeps all her records in a paper notebook and wants to move them into a database.
>
> The bank has customers. For each customer, Mere records a customer ID, first name, last name and phone number. Each customer has a different customer ID.
>
> Customers open accounts. Each account has an account number, an account type (savings or everyday) and a balance. Each account has a different account number.
>
> A customer owns one or more accounts. Each account belongs to only one customer.

Questions

1. List the entities. There are two.
2. List the attributes of each entity and identify the key attribute.
3. Identify the relationship and its cardinality.
4. Draw the ER diagram using Chen's notation.

Hint: *Look for nouns that the bank stores many of. The bank itself is not an entity, because it is the whole system.*

Working space: entities, attributes and relationships

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TASK 1 Simple · Bank · About 10 minutes

Draw your ER diagram

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TASK 2 Simple · University · About 10 minutes

Students and clubs

Two entities and a many-to-many relationship. One attribute belongs to the relationship.

Scenario

> Sione works in the student centre at Southfield University. He looks after the student clubs, such as the chess club, the hiking club and the cooking club.
>
> Each student has a student ID, first name, last name and email. Each club has a club ID, club name and meeting day.
>
> A student can join many clubs, and a club has many students. Sione also wants to record the date each student joined each club.

Attributes on a relationship

Some facts do not belong to either entity on its own. The join date is one of them. A student in three clubs has three different join dates, and a club has a different join date for each member. The date only makes sense for one student and one club together, so it is connected to the relationship diamond.

Questions

1. List the entities.
2. List the attributes of each entity and identify the key attribute.
3. Identify the relationship and its cardinality.
4. Decide where the join date belongs.
5. Draw the ER diagram using Chen's notation.

Hint: *Ask two questions. How many clubs can one student join? How many students can one club have?*

Working space: entities, attributes and relationships

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TASK 2 Simple · University · About 10 minutes

Draw your ER diagram

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TASK 3 Moderate · Hospital · About 20 minutes

Wards, nurses, doctors and patients

Four entities and three relationships. Work on one relationship at a time.

Scenario

> Priya is a charge nurse at Riverside Hospital. The hospital tracks patients on whiteboards, and Priya has been asked to help design a database.
>
> The hospital has wards. Each ward has a ward number, ward name and number of beds.
>
> Patients stay in wards. Each patient stays in one ward, and a ward has many patients. For each patient, the hospital records a patient ID, first name, last name and date of birth.
>
> Nurses work in wards. Each nurse works in one ward, and a ward has many nurses. A nurse has a nurse ID, first name, last name and shift (day or night).
>
> Doctors treat patients. A doctor treats many patients, and a patient can be treated by many doctors. A doctor has a doctor ID, first name, last name and specialty. Each time a doctor treats a patient, the hospital records the treatment date and the diagnosis.

Questions

1. List the entities. There are four.
2. List the attributes of each entity and identify the key attributes.
3. Identify the three relationships and the cardinality of each.
4. Which relationship has its own attributes?
5. Draw the ER diagram using Chen's notation.

Hint: *Draw the four entities first and leave space between them. Then add one relationship at a time and work out its cardinality before moving to the next.*

Working space: entities, attributes and relationships

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TASK 3 Moderate · Hospital · About 20 minutes

Draw your ER diagram

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TASK 4 Moderate · Bank · About 25 minutes

Branches, employees and joint accounts

The bank from Task 1 with new rules. Two entities are connected by two different relationships.

Scenario

> This task uses Mere's bank from Task 1. The bank now has three branches and needs a larger database.
>
> The bank has branches. Each branch has a branch code, branch name and city.
>
> Customers have a customer ID, first name, last name and phone number, as before. Accounts have an account number, account type and balance.
>
> One rule has changed. The bank now offers joint accounts, so a customer can own many accounts and an account can be owned by more than one customer. For example, two partners can share one account. The bank records the date each customer was added to an account.
>
> Each account is opened at one branch. A branch holds many accounts.
>
> The bank also has employees. Each employee has an employee ID, first name, last name and job title. Each employee works at one branch, and a branch has many employees. Each branch is managed by one employee, and an employee can manage only one branch.

Questions

1. List the entities. There are four.
2. List the attributes of each entity and identify the key attributes.
3. Identify the four relationships and the cardinality of each.
4. In Task 1, OWNS was 1:N. What is it now, and why?
5. EMPLOYEE and BRANCH are connected in two ways. Show both relationships.
6. Draw the ER diagram using Chen's notation.

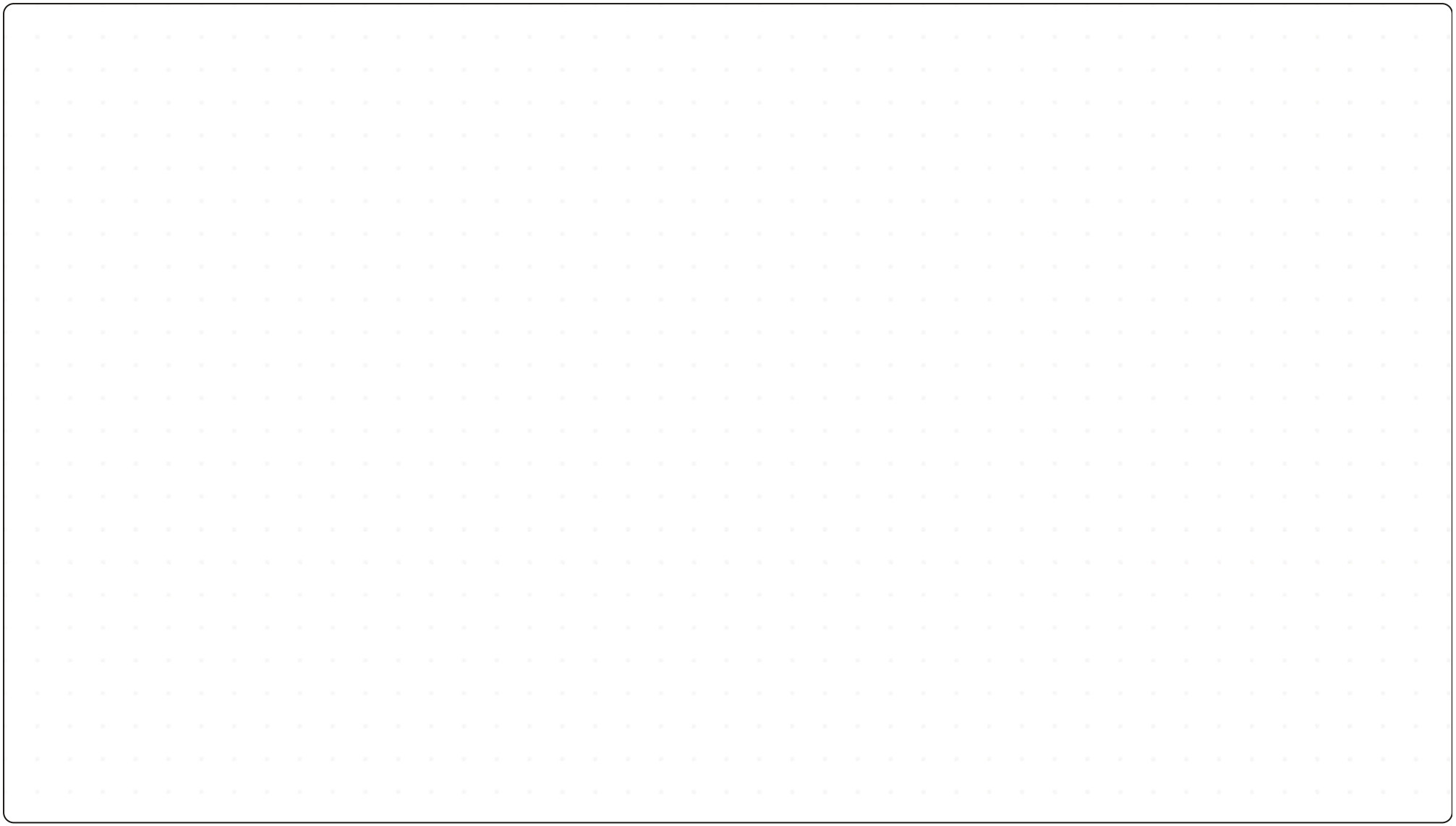
Hint: *Working at a branch and managing a branch are two different facts. Each one needs its own relationship, even though both connect EMPLOYEE and BRANCH.*

Working space: entities, attributes and relationships

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TASK 4 Moderate · Bank · About 25 minutes

Draw your ER diagram



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TASK 5 Moderate · University · About 25 minutes

Courses, lecturers and assignments

Four entities and four relationships. Three attributes belong to relationships.

Scenario

> Dr Aroha Ngata teaches database design at Southfield University. She wants a database for her courses and their assignments.
>
> Each lecturer has a staff ID, first name, last name and email. A lecturer teaches many courses, and each course is taught by one lecturer.
>
> Each course has a course code, title and number of credits.
>
> Each student has a student ID, first name, last name and email. A student takes many courses, and a course has many students. Each student receives a grade for each course they take.
>
> Each course sets several assignments, and each assignment belongs to one course. An assignment has an assignment ID, title, due date and weight (the percentage it counts towards the course).
>
> Students submit assignments. A student submits many assignments, and each assignment is submitted by many students. For each submission, the university records the date submitted and the mark.

Questions

1. List the entities. There are four.
2. List the attributes of each entity and identify the key attributes.
3. Identify the four relationships and the cardinality of each.
4. Which relationships have their own attributes?
5. Draw the ER diagram using Chen's notation.

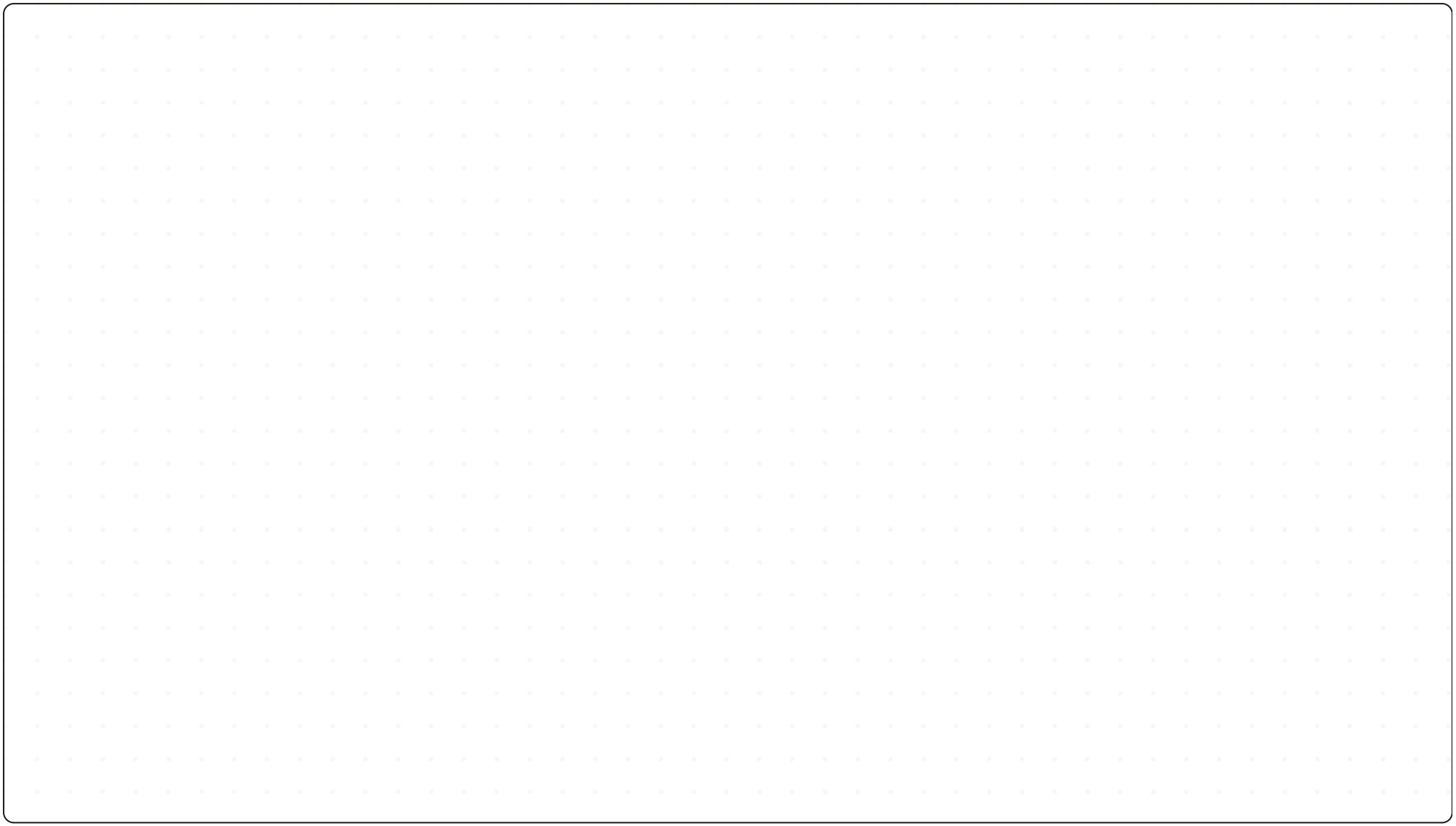
Hint: *Look at Grade. Does it describe a student, a course, or one student in one course? Ask the same question about Mark.*

Working space: entities, attributes and relationships

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TASK 5 Moderate · University · About 25 minutes

Draw your ER diagram



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TASK 6 Simple · Online shop · About 5 minutes

Customers and purchases

A very small scenario about one new idea: must every instance take part in the relationship?

Scenario

> Hemi sells handmade candles online. Each customer has a customer ID and a name. Each purchase has a purchase number and a purchase date.
>
> A customer makes purchases, and can make many. A customer can register on the website before buying anything. Every purchase must be made by exactly one customer.

Participation: total and partial

Cardinality counts how many. Participation asks whether an entity must take part at all. Total participation means every instance must be in the relationship, so it is drawn with a double line. Partial participation means some instances may not be in it, so it is drawn with a single line. Ask the question for each entity separately.

Questions

1. List the entities and the attributes of each, and identify the key attributes.
2. Identify the relationship and its cardinality.
3. Must every CUSTOMER take part in the relationship? Must every PURCHASE? Decide whether each side is total or partial.
4. Draw the ER diagram using Chen's notation. Show total participation with a double line.

Hint: *For each entity, ask: can one exist without being in the relationship? If it can, the participation is partial. If it cannot, it is total.*

Working space: entities, attributes and relationships

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TASK 6 Simple · Online shop · About 5 minutes

Draw your ER diagram

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PART 3

Checklist and common mistakes

Before you check your answers

Go through each of your diagrams and tick each point.

- [ ] Every entity is a singular noun in capitals, inside a rectangle.
- [ ] Every entity has one key attribute, and its name is underlined.
- [ ] Every relationship is a verb in capitals, inside a diamond, connected to two entities.
- [ ] Every relationship line has a 1, N or M next to its entity.
- [ ] Every ellipse is connected to an entity or a relationship.

Common mistakes

Mistake	How to fix it
Drawing the organisation as an entity	BANK or HOSPITAL is the whole system, not one entity in it. Leave it out.
Using plural names	Write CUSTOMER, not CUSTOMERS. An entity name describes one instance.
Leaving out the key	Every entity needs one key attribute, underlined. If the scenario has no unique attribute, add an ID.
Putting a relationship attribute on an entity	If a fact needs both entities to make sense, such as Grade, attach it to the relationship.
Guessing the cardinality	Use the rules in the scenario. In Task 1 each account has one owner, even though real banks often allow joint accounts.
Using a noun for a relationship	Write OWNS or TAKES, not OWNERSHIP or ENROLMENT.

Your layout can be different from the answer. Your diagram is correct if it has the same entities, keys, relationships and cardinalities.

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